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section .data
    m1 db      10,"Enter 4-digit Hex number: "
    l1 equ      $- m1
    m2 db 10,"Equivalent BCD number is: "
    l2 equ      $- m2
    emsg db 10,"You entered Invalid Data!!!",
    l3 equ      $-emsg
;-----
section .bss
    buf          resb  6
    buf_len:     equ   $-buf
    digitcount   resb  1
    ans          resw  1
    char_ans     resb  4
;-----
;macros as per 64-bit conversions

%macro print 2
    mov  rax,1      ; Function 1 - write
    mov  rdi,1      ; To stdout
    mov  rsi,%1     ; String address
    mov  rdx,%2     ; String size
    syscall          ; invoke operating system to WRITE
%endmacro

%macro read 2
    mov  rax,0      ; Function 0 - Read
    mov  rdi,0      ; from stdin
    mov  rsi,%1     ; buffer address
    mov  rdx,%2     ; buffer size
    syscall          ; invoke operating system to READ
%endmacro

%macro exit 0
    mov  rax, 60    ; system call 60 is exit
    mov  rdi,0      ; we want return code 0
    syscall          ; invoke operating system to exit
%endmacro
;-----
section .text
    global _start
_start:
    print  m1,l1
    call  accept_num
    mov   ax,bx

    mov   rbx,10
back:
    xor   rdx,rdx
    div   rbx

    push  dx          ;Remainder

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        inc     byte[digitcount]

        cmp     rax,0h           ;quotient
        jne     back

        print   m2, l2

print_bcd:
        pop     dx
        add     dl,30h           ; possible digits are 0-9 so add 30H only
        mov     [char_ans],dl ; store character in char_ans

        print   char_ans,1      ; print on screen in reverse order

        dec     byte[digitcount]
        jnz     print_bcd

        exit

;-----
accept_num:
        read    buf,5           ; buflen = 4 + 1

        xor     bx,bx
        mov     rcx,4
        mov     rsi,buf

next_digit:
        shl     bx,04
        mov     al,[rsi]

        cmp     al,"0"          ; "0" = 30h or 48d
        jb     error            ; jump if below "0" to error
        cmp     al,"9"
        jbe     sub30           ; subtract 30h if no is in the range "0"-"9"

        cmp     al,"A"          ; "A" = 41h or 65d
        jb     error            ; jump if below "A" to error
        cmp     al,"F"
        jbe     sub37           ; subtract 37h if no is in the range "A"-"F"

        cmp     al,"a"          ; "a" = 61h or 97d
        jb     error            ; jump if below "a" to error
        cmp     al,"f"
        jbe     sub57           ; subtract 57h if no is in the range "a"-"f"

error:  print   emsg, l3        ; "You entered Invalid Data!!!"
        exit

sub57:   sub     al,57h         ; subtract 57h if no is in the range "a"-"f"
        jmp     next
sub37:  sub     al,37h         ; subtract 37h if no is in the range "a"-"f"
        jmp     next
sub30:  sub     al,30h         ; subtract 30h if no is in the range "0"-"9"

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next: add    bx,ax      ; prepare number
      inc    rsi        ; point to next digit
      loop   next_digit
ret
```